



NAMIBIA UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Database Fundamentals (DBF510S) Course Outline

Faculty Name: Computing and Informatics

Name of Department: Software Engineering

Effective Start Date: 09 February 2026

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

All staff and students of the Namibia University of Science and Technology (NUST), upon signing their employment contracts and registration forms, commit themselves to abide by the policies and rules of the institution. The core activity of NUST is learning and in this respect, academic honesty and integrity are very important to ensure that learning is valid, reliable, and credible.

NUST, therefore, does not condone any form of academic dishonesty, including plagiarism and cheating on tests and assessments, amongst other such practices. NUST requires students to always do their assignments and to produce their academic work unless given a group assignment.

Academic Dishonesty includes, but is not limited to:

- Using the ideas, words, works, or inventions of someone else as if it is your work.
- Using the direct words of someone else without quotation marks, even if it is referenced.
- Copying from writings (books, articles, webpages, other students' assignments, etc.), published, or unpublished, without referencing.
- Syndication of a piece of work, all or part of an assignment, by a group of students, unless the assignment was a legitimate group assignment.
- The borrowing and use of another person's assignment, with or without their knowledge or permission.
- Infringing copyright, including documents, copied or cut and pasted from the internet.
- Asking someone else to prepare an assignment for you or to write or sit an assessment for you, whether this is against payment or not.
- Re-submitting work done already for another course or programme as new work is called self-plagiarism.
- Bringing notes into an examination or test venue, regardless of whether the notes were used to copy or not.
- Receiving any outside assistance in any form or shape during an examination or test.

All forms of academic dishonesty are viewed as misconduct under NUST Student Rules and Regulations. Students who make themselves guilty of academic dishonesty will be brought before a Disciplinary Committee and may be suspended from studying for a certain time or may be expelled. All students who are found guilty of academic dishonesty shall have an appropriate endorsement on their academic record, which will never be erased.

1. Course Overview

- **Course Code and Title:** DBF510S, Database Fundamentals
- **Department:** Software Engineering
- **Programmes:**
 - 07BOAI, BACHELOR OF ARTIFICIAL INTELLIGENCE
 - 07BCMS, BACHELOR OF COMPUTER SCIENCE NQF: 7,
 - 07BCSS, BACHELOR OF COMPUTER SCIENCE,
 - 07BCCY, B.COMP. SC. IN CYBER SECURITY NQF: 7,
 - 07BAIT, BACHELOR OF INFORMATICS NQF: 7,
 - 07BAIN, BACHELOR OF INFORMATICS NQF: 7,
 - 07BGEI, B.: GEOINFORMATION TECHNOLOGY NQF: 7,
 - 07BLAM, BACHELOR OF LAND ADMINISTRATION NQF: 7,
 - 07BLAN, BACHELOR OF LAND ADMINISTRATION NQF: 7
- **Notational Hours:** 100 hours (Contact: 45 hours; Directed self-learning and Self-Directed learning: 40 hours; Assessment: 15 hours)
- **NQF Level and Credit:** Level 5, 10 Credits
- **Semester Offered:** 1
- **Pre-Requisites:** None
- **Options:** Compulsory

2. Course Aims

The aim of this course is to equip students with the knowledge and skills needed to design and develop a database using models, techniques and standard development processes. It also exposes the students to the knowledge and skills required in the implementation of database systems using Structured Query Language (SQL).

3. Learning Outcome

Specific Outcomes:

On successful completion of the course students should be able to show evidence of their ability to:

- Evaluate the role and the concepts of a database system.
- Translate requirements into conceptual designs.
- Design logical databases.
- Formulate SQL queries on databases.

Comprehensive Learning Outcome:

Design relational, logical, physical databases and applications, using models, techniques and standard development processes.

4. Course Content and Weekly Schedule

(Refer to 2026 NUST Institutional Calendar for detailed dates)

Course Schedule

Week #	Dates	Topics and Subtopics	Activities, Discussion Forums, Reading Materials and Assessments
Week 1	9/02/2026- 13/02/2026	Course Introduction -course outline -software -learning platforms	Set up the labs and user accounts
Week 2	16/02/2026- 20/02/2026	The Database Environment -History of database applications -Characteristics of Database Approach -Advantages of the DBMS approach -Database and database users -An example database	Discussion Forum: The Role of Database Management Systems Activity: Lab 1
Week 3	23/02/2026- 27/02/2026	Database System Concepts & Architecture -Types of database systems -database system concepts -Relational Model	Activity: Lab 1
Week 4	02/03/2026- 06/03/2026	The Database Development Process	Activity: Lab 2

Week 5	09/03/2026- 13/03/2026	Database Design -conceptual data model -Entity Relationship Model	Activity: Lab 3 Entity Relationship Model
Week 6	16/03/2026- 20/03/2026	Normalisation and Functional Dependency	Activity: Lab 4 Entity Relationship Model
Week 7	23/03/2026- 27/03/2026	Database design -logical database schema -Understanding the difference between Logical Database Schema and Physical Database Schema.	Activity Lab 5 Discussion Forum: Normalisation and Functional Dependency
Week 8	30/03/2026- 02/04/2026	Database design Continued	
	03/04/2026-06/04/2026	RECESS	
Week 9	07/04/2026-10/04/2026	Introduction to SQL and Data Definition Language	Activity Lab 6
Week 10	13/04/2026-17/04/2026	Introduction to SQL and Data Definition Language	
Week 11	20/04/2026-24/04/2026	Data Manipulation Language Data Control Language	Activity Lab 7
Week 12	27/04/2026-30/04/2026	Data Manipulation Language Data Control Language	Activity Lab 8
Week 13	05/05/2026-08/05/2026	Revision	
Week 14	11/05/2026-15/05/2026	Revision	
Week 15	18/05/2026	Last Day for Changing Semester Marks	

5. Delivery Methods

Course Delivery Methods:

The course will be facilitated through lectures, presentations, discussions, case studies, theory and practical demonstrations, self-guided learning, assignments, and exercises.

Lectures - this will present the foundation theories of the course. Students are required to deepen their knowledge and understanding of the covered subjects through guided self-study and laboratory sessions.

Laboratory activities – this will be an avenue for the students to try out and acquire practical skills of theoretical knowledge gained. Laboratory materials will be prepared with a clear indication of what the student is expected to achieve, a problem statement, and guidelines on how to approach the problem. The following communication tools will be used in this course:

Moodle (elearning), Microsoft Teams, and email.

Course format: All modes of delivery and participation will make use of technology-enriched learning through eLearning. The learning process will be facilitated both in and outside the classroom, requiring specific tasks to be carried out by the student. This facilitation will make use of practical projects, tutorials, case studies, problem-based learning assignments, individual work, group work, classroom observation and feedback, and creative and critical thinking. The progress of learning embedded in such tasks will be monitored, recorded, and assessed through tests, quizzes, and assignments.

15 contact hours of directed learning (1 Theory lecture weekly).

30 hours of directed learning (2 laboratory practical sessions weekly).

15 hours of self-directed learning (1 practical self-study session weekly)

25 hours of self-directed learning to be used for reading prescribed material, preparing assignments/projects, homework & tests.

Assessments: 15 hours (tests, assignment & home-works presentations and examination).

Quality Assurance Arrangements:

Moderation of the assessment will be done in accordance with the NUST's general rules and guidelines on moderation. Lecturers will utilise self-evaluation outcomes to improve the quality of learning materials and facilitation methods. Internal moderation will be required to foster the quality of materials and delivery. Students' evaluation of course content and delivery methods will be conducted at regular intervals.

6. Student and Institutional Support

Lecturers Information

Instructor	Role	Office location	Email	Office phone
Mrs Shilumbe Chivuno-Kuria	Theory and Practical Lecturer Course Coordinator	R.404, 4 th Floor Poly Heights.		2072057
Mr Andrew Tjirare	Practical Lecturer	IT House		
Ms Laurinda Ndjao	Practical Lecturer	IT House		
Ms Tjihimise Kaunatjike	Practical Lecturer	IT House		
Mr Keith Kasika	Practical Lecturer	IT House		
Mr Nyasha Musiyarira	Practical Lecturer	IT House		
Ms Minikuee Kasaona	Practical Lecturer	IT House		

The practical lecturers will inform the practical groups of their contact details, consultation hours and modes of communication.

Office Hours: Course Coordinator and Lecturers' office hours will be posted on E-Learning or their office doors.

7. Course Requirements for Students

Technology & Equipment Readiness: This course will only utilise already existing NUST equipment.

Student Commitments and Contact Times:

a) **Class Participation:** It is advised that every student in the class participate in discussions such as making comments, asking questions, or responding to questions asked by the instructor or other students. Discussions about the lesson will help you to understand the topic better and to see others' perceptions and views on a topic and it gives opportunity to ask questions.

- b) **Working in Groups:** Class exercises or activities will be done in groups as collaborating with fellow students will help you discover your strengths and weaknesses. Furthermore, working in a team will develop your team working skills such as leadership, working with and motivating others as well as interpersonal skills in a way that will build your self-confidence to be able to speak with boldness and listen to others' ideas.
- c) **Class Attendance:** Attend all classes, be on time, and make appropriate notes. If a student fails to attend a class, it is their responsibility to catch up on what was covered. The students have access to the recommended books from the library. If any special presentations are used in class, they will be made available and do not need to be copied by the student.
- d) **Self-Directed study:** After every class, and before the start of the next class, sort, complete, and annotate your lecture notes, and select further reading in case any particular item was not understood on the first attempt. Student must spend their notational hours revising, practicing, or/and completing homework for the course. Students should take the reading list (prescribed and recommended reading seriously).
- e) **Communications:** Maintain active communication with your group lecturer. Students are expected to use established channels of communication and platforms to solve course-related issues on time. The end of the semester is not the best time to fix issues. Use the available time to deal with matters at hand.

Course Resources

There are several facilities that students can use as a course resource at NUST and at home.

- a) **NUST Level:** The main library, Department of Student Services, Writing centre and student academic problems - Teaching & Learning Unit (TLU), Campus Health and Wellness Centre (CHWC) - NUST Clinic, Early alert staff.
- b) **Faculty Level:** Faculty-based induction, learning support units, and tutor systems, computer laboratories. Lecturers are available for consultation and the hours are shown on their doors. Emails can be sent to the lecturers as well.
- c) **Course Level:** Course Outline; Electronic learning resources in the course folder on E-learning. Make use of the MyNUST eLearning platform as the learning management application.

Prescribed Textbook:

Coronel, C. (2022). *Database System: Design, Implementation, & Management*. (14th Ed.). Cengage Learning. ISBN-13-978- 0357673034

Silberschatz A., Korth H., & Sudarshan S. (2019). *Database System Concepts*. (7th Ed.). McGraw – Hill, ISBN-13-978-0078022159

Recommended Readings:

Elmasri, R. (2022). *Fundamentals of Database Systems*, (7th ed.). Generic.

Shields, W. (2019). *SQL Quickstart Guide: The Simplified Beginner's Guide to Managing, Analysing, and Manipulating Data With SQL (Coding & Programming – Quickstart Guides) Systems: the complete Book. Coding & Programming Quickstart Guides*. ISBN-13-978-1945051753

8. Assessment and Evaluation:

The course will be assessed using diversified Continuous Assessments and a final examination with feedback only (the final determination of the weightings shall be done by the course examiner):

A minimum Final Mark of **50%** is required to pass the course.

Assessment Tasks Weighting

Assessment	Weight	Date	Venue
Quiz 1	10%	12-13 March 2026	e-learning
Quiz 2	10%	1-2 April 2026	e-learning
Quiz 3	10%	7-8 May 2026	e-learning
Lab X	10%	TBD	Submission on e-learning
Lab Y	10%	TBD	Submission on e-learning
Test 1	50%	10 April 2026	TBA

Make up Test		17 April 2026	TBA
Supplementary Test		30 April 2026	TBA

General Academic Policies:

It is the student's responsibility to be familiar with and adhere to the NUST's Policies.

All Late Assignments will not be accepted after the due date and time and thus be marked as 0%.

Failure to Pay Fees:

A student who fails to pay his/her fees may not be allowed to write the examination and if allowed, the results will be withheld until all outstanding fees are paid in full.

Authorisation:

This course is authorised for use by:



Head of Department

17 February 2026

Date

ACKNOWLEDGEMENT BY STUDENT

(To be completed by all students on the course, detached from the course outline, and kept on record in the department)

I, _____, student Number _____, hereby acknowledge that I have received this course outline for Database Fundamentals (DBF510S) in semester 1 (2026), and that I have familiarised myself with its content, in particular the statement about academic honesty and integrity. I agree to abide by the Policies and arrangements spelt out in this course outline.

Signature of student

Date